Golden-baseline reference scenario, resolved settings

Companion to Appendix C of the thesis.
Generated from `config/golden_baseline.json`.

The settings below are the resolved contents of `config/golden_baseline.json`, excepting eight that name output files and set diagnostic storage and so cannot reach any reported quantity. Two cautions carry with the listing. First, the file is an overlay, so it states only what differs from the master configuration and every setting not listed here takes its master default. Second, a setting appearing here means the scenario file declares it, not that it necessarily reaches the physics. The framework carries keys that no longer have a consumer, and a listing cannot by itself distinguish a live setting from an inert one. Claims in the thesis rest on measured runs and on the regression gates, never on the presence of a key in a configuration file.

Table 1: Settings of the golden-baseline reference scenario

Setting	**Short description**	Value
Scenario and run		
`scenario.nReceivers`	Number of receive antenna phase centres on the spacecraft	`4`
`scenario.nSpaceAssets`	Number of space assets simulated	`1`
`scenario.nTowers`	Number of ground tracking stations in the network	`5`
`scenario.name`	Scenario name label used in reports and run fingerprints	`golden_baseline`
`scenario.orbitClass`	Orbit class of the tracked spacecraft	`GEO`
`simulation.dt_s`	Measurement and propagation time step in seconds	`1`
`simulation.seed`	Master random number seed for bit-reproducible runs	`42`
Grade overlays		
`atmosphere.estimateIono`	Profile shortcut enabling slant ionosphere states, dropping Klobuchar correction	`false`
`atmosphere.ionosphereFree`	Selects ionosphere-free code combination rather than raw dual frequency	`false`
`atmosphere.realistic`	Selects the physically realistic troposphere and ionosphere profile	`true`
`atmosphere.sharedAcrossAntennas.enable`	Shares one scintillation draw across all receive antennas	`true`
`realism.grade`	Opt-in switch for the whole realism error overlay	`true`
`realism.include.interAntennaCarrierBias`	Overlay sub-toggle for per-antenna carrier line bias	`true`
`realism.include.luniSolar`	Overlay sub-toggle for the reduced-dynamics luni-solar stress case	`false`
`realism.include.solidEarthTide`	Overlay sub-toggle for truth-only solid Earth tide displacement	`false`
Orbit and force model		
`orbit.truth.perturbations.epochJD_TT`	Sun and Moon ephemeris reference epoch, Julian date	`2451545.0`
`orbit.truth.perturbations.luniSolar.enable`	Sun and Moon third-body acceleration in truth propagation	`true`
`orbit.truth.perturbations.srp.Cr`	Cannonball radiation pressure reflectivity coefficient, dimensionless	`1.3`
`orbit.truth.perturbations.srp.areaToMass_m2pkg`	Spacecraft area-to-mass ratio, square metres per kilogram	`0.02`
`orbit.truth.perturbations.srp.enable`	Solar radiation pressure acceleration in truth propagation	`true`
`orbit.truth.perturbations.srp.shadow`	Eclipse shadow geometry used for radiation pressure	`cylindrical`
Propagation and geometry effects		
`effects.antenna.pcvModel`	Antenna phase centre variation model, applied to truth and model	`toy`

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Setting	**Short description**	Value
effects.antennaPCO.calibration Residual.enable	Gate for truth-only phase centre offset mis-calibration	true
effects.antennaPCO.calibration Residual.receiverOffset_body_m	Receive phase centre mis-calibration per body axis, metres	[0.002,0.002,0.002]
effects.ionosphere.mappingModel	Obliquity mapping converting vertical to slant ionospheric delay	thinShell
effects.ionosphere.shellHeight_m	Ionospheric thin-shell pierce point height, metres	350000
effects.solidEarthTide.truth.enable	Displaces true tower positions by the solid Earth tide	false
effects.towerSurvey.enable	Master gate for the tower coordinate survey error	true
effects.towerSurvey.model.enable	Applies the survey error to the estimator's tower positions	false
effects.towerSurvey.sigmaENU_m	Tower survey error one-sigma, east north up, metres	[0.01,0.01,0.03]
effects.towerSurvey.truth.enable	Applies the survey error to the true tower positions	true
frames.eopModel.enable	Applies published IERS orientation correction inside the estimator	true
frames.eopModel.polarMotion_xp_ arcsec	Predicted pole x used by the estimator, arcseconds	0.1497
frames.eopModel.polarMotion_yp_ arcsec	Predicted pole y used by the estimator, arcseconds	0.3497
frames.eopModel.ut1Rate_error_msPer Day	Predicted length-of-day rate correction, milliseconds per day	0.04
frames.truthEop.enable	Applies truth-side Earth orientation error to tower positions	true
frames.truthEop.polarMotion_xp_ arcsec	True pole x displacement, arcseconds	0.15
frames.truthEop.polarMotion_yp_ arcsec	True pole y displacement, arcseconds	0.35
frames.truthEop.ut1Rate_error_msPer Day	True length-of-day rate error, milliseconds per day	0.05
physics.relativity.clock.enable	Master gate for the relativistic clock rate offset	true
physics.relativity.clock.model. enable	Estimator models the relativistic clock rate offset	true
physics.relativity.clock.truth. enable	Puts the relativistic clock rate offset in truth	true
Clocks		
clock.gauge.mode	How the clock datum ambiguity is resolved	externalTowerCorrections
clock.mode	Which clocks enter the EKF state vector	spacecraftReceiverClockOnly
clocks.tower.product.addToR	Charges the product uncertainty to measurement noise covariance	true
clocks.tower.product.latency_s	Delivery latency of the clock correction, seconds	5
clocks.tower.product.sigmaBias_m	Broadcast tower clock bias residual sigma, metres	0.1
clocks.tower.product.sigmaDrift_mps	Broadcast clock drift residual sigma, metres per second	0.001
clocks.tower.product.update Interval_s	Rebroadcast interval of the clock correction product, seconds	30
clocks.tower.product.validity_s	Validity span of one clock correction, seconds	120
Truth-side error sources		
biases.interFrequency.code.model. L1_m	L1 code bias the estimator corrects, metres	0.3
biases.interFrequency.code.model. L2_m	L2 code bias the estimator corrects, metres	0.405
biases.interFrequency.code.truth. L1_m	True L1 code group delay bias, metres	0.3
biases.interFrequency.code.truth. L2_m	True L2 code group delay bias, metres	0.45
errors.hardwareDelay.enable	Master switch for the tower hardware group delay	true

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Setting	**Short description**	Value
errors.hardwareDelay.model.enable	Applies a hardware delay correction on the estimator side	false
errors.hardwareDelay.residualStochastic.enable	Draws the white hardware delay residual and charges R	true
errors.hardwareDelay.sigma_m	Hardware delay calibration residual one sigma in metres	0.05
errors.hardwareDelay.truth.enable	Adds the hardware delay to the truth range	true
errors.interAntennaCarrierBias.drift.enable	Adds thermal drift to the inter-antenna carrier bias	false
errors.interAntennaCarrierBias.enable	Injects an unknown per-antenna carrier phase line bias	true
errors.interAntennaCarrierBias.perSignal	Draws an independent bias for each carrier signal	true
errors.interAntennaCarrierBias.sigma_cycles	Inter-antenna carrier phase bias one sigma in cycles	0.25
errors.ionosphere.enable	Master switch for the ionospheric delay error source	true
errors.ionosphere.higherOrder.enable	Add second and third order ionosphere residuals, truth only	true
errors.ionosphere.model.correction	Which broadcast ionosphere correction the estimator applies	klobuchar
errors.ionosphere.model.enable	Apply the estimator side ionospheric correction	true
errors.ionosphere.modelType	Ionosphere formulation, unlocking the stochastic TEC branch	tecGaussMarkov
errors.ionosphere.scintillation.S4zen	Zenith amplitude scintillation index, dimensionless	0.3
errors.ionosphere.scintillation.enable	Add ionospheric amplitude scintillation to truth and R	true
errors.ionosphere.scintillation.model	Scintillation formulation supplying the fading factor	conker
errors.ionosphere.scintillation.phaseScint.enable	Add phase scintillation to the truth carrier	true
errors.ionosphere.scintillation.phaseScint.sigmaPhi_rad	Phase scintillation standard deviation, radians	0.2
errors.ionosphere.scintillation.phaseScint.tau_s	Correlation time of the phase scintillation, seconds	1.5
errors.ionosphere.scintillation.tau_s	Correlation time of the scintillation amplitude, seconds	30
errors.ionosphere.sigma_m	Declared vertical ionosphere uncertainty charged in R, metres	1.0
errors.ionosphere.stochastic.enable	Add a time correlated residual TEC component	true
errors.ionosphere.stochastic.sigmaVDelayL1_ss_m	Steady state vertical L1 residual delay, metres	0.3
errors.ionosphere.stochastic.tau_s	Correlation time of the residual TEC process, seconds	600
errors.ionosphere.topsideFraction	Fraction of the vertical TEC column the spacecraft sees	1
errors.ionosphere.truth.diurnal.enable	Vary the truth vertical TEC over the daily cycle	true
errors.ionosphere.truth.diurnal.peakLocalTime_h	Local solar time of the TEC peak, hours	14
errors.ionosphere.truth.diurnal.vtecDay_TECU	Daytime peak vertical TEC, TECU	30
errors.ionosphere.truth.diurnal.vtecNight_TECU	Night-time floor vertical TEC, TECU	6
errors.ionosphere.truth.enable	Inject ionospheric delay into the simulated truth observations	true
errors.multipath.coloredGM.elevationExponent	Exponent of the inverse-sine elevation multipath envelope	1
errors.multipath.coloredGM.enable	Selects the coloured Gauss-Markov multipath process over the white model	true
errors.multipath.coloredGM.sharedAcrossAntennas.enable	Shares one multipath state per tower across all antennas	true

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Setting	**Short description**	Value
errors.multipath.coloredGM.sigmaCodeL1_ss_m	Steady-state L1 code multipath one sigma in metres	0.3
errors.multipath.coloredGM.tau_s	Multipath correlation time in seconds	60
errors.multipath.enable	Master switch for the code multipath error	true
errors.multipath.model.enable	Applies a multipath correction on the estimator side	false
errors.multipath.truth.enable	Adds multipath to the truth pseudorange	true
errors.troposphere.dayOfYear	Day of year driving the Niell mapping seasonal term	180
errors.troposphere.enable	Master switch for the tropospheric delay error source	true
errors.troposphere.model.enable	Apply the estimator side tropospheric delay correction	true
errors.troposphere.model.mappingType	Elevation mapping function used by the estimator model	niell
errors.troposphere.modelType	Troposphere formulation, surface weather hydrostatic plus stochastic wet delay	localWeatherGM
errors.troposphere.sigma_m	Declared zenith troposphere uncertainty charged in R, metres	0.15
errors.troposphere.stochastic.enable	Add a time correlated residual zenith wet delay	true
errors.troposphere.stochastic.sigmaWet_ss_m	Steady state zenith wet delay variability, metres	0.04
errors.troposphere.stochastic.tau_s	Correlation time of the wet delay process, seconds	10800
errors.troposphere.truth.enable	Inject tropospheric delay into the simulated truth observations	true
errors.troposphere.truth.mappingType	Elevation mapping function used on the truth side	niell
Measurements		
carrierSlip.baselineDifferencedMode.enable	Whether slip metrics are differenced against a reference antenna	true
carrierSlip.baselineDifferencedMode.referenceAntenna	Index of the antenna used as differencing reference	1
carrierSlip.commonModeCompensation.enable	Whether the common-mode jump is removed before slip detection	true
carrierSlip.commonModeCompensation.minRows	Minimum valid carrier rows needed to estimate the common jump	4
measurement.sigmaFloor_m	Lower bound applied to any measurement sigma, metres	0.01
measurements.carrier.sigma_m	Carrier phase measurement noise sigma, metres	0.01
measurements.carrierMode	Carrier phase mode, off, diagnostic or float ambiguity	ekfFloat
measurements.carrierPhase.enable	Legacy carrier phase gate, a fallback when carrier mode absent	true
measurements.codeNoise.cn0.base_dBHz	Baseline carrier-to-noise density in the C/N0 link model, dB-Hz	45
measurements.codeNoise.cn0.elevationGain_dB	Carrier-to-noise density gain from horizon to zenith, decibels	6
measurements.codeNoise.cn0.sigmaAt45dBHz_m	Reference code noise sigma at 45 dB-Hz, metres	0.3
measurements.codeNoise.model	Code noise model, constant, elevation weighted or C/N0	cn0
measurements.doppler.enable	Whether Doppler range-rate observables are generated	true
measurements.doppler.sigma_mps	Doppler measurement noise sigma, metres per second	0.0424
measurements.doppler.useInEKF	Whether Doppler rows update the filter	true
measurements.twoWayTimeTransfer.conservativeProductCorrelation	Whether tower clock product variance is inflated for within-interval correlation	true
measurements.twoWayTimeTransfer.enable	Whether tower-spacecraft two-way time transfer rows are generated	true

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Setting	**Short description**	Value
`measurements.twoWayTimeTransfer.includeReciprocityResidual`	Whether motion non-reciprocity of the two-way path is modelled	`true`
`measurements.twoWayTimeTransfer.mode`	Two-way observable formulation, first-order reciprocal or four-timestamp	`firstOrderReciprocal`
`measurements.twoWayTimeTransfer.reciprocitySigma_m`	Residual non-reciprocity sigma of the two-way path, metres	`0.005`
`measurements.twoWayTimeTransfer.sigma_m`	Two-way time transfer noise sigma, metres	`0.03`
`measurements.twoWayTimeTransfer.towers`	Which ground towers are two-way capable	`all`
`measurements.twoWayTimeTransfer.useInEKF`	Whether two-way time transfer rows update the filter	`true`
`measurements.twoWayTimeTransfer.warmup_s`	Delay before two-way rows begin, seconds	`0`
Estimator configuration		
`estimation.ambiguity.initialSigma_m`	Initial one-sigma prior on each carrier ambiguity state, metres	`100`
`estimation.ambiguityMode`	How carrier phase ambiguities are parameterised as filter states	`floatPerTowerReceiverSignal`
`estimation.ionosphereMode`	Whether and how the filter estimates ionospheric delay states	`perTowerSlant`
`estimation.slantIono.initialSigma_m`	Initial one-sigma prior on the slant-ionosphere state, metres	`5.0`
`estimation.slantIono.sigma_ss_m`	Steady-state sigma of the slant-ionosphere state, metres	`1.0`
`estimation.slantIono.tau_s`	Correlation time of the slant-ionosphere Gauss-Markov state, seconds	`600`
`estimation.tropoZwd.initialSigma_m`	Initial one-sigma prior on the zenith wet delay state, metres	`0.1`
`estimation.tropoZwd.sigma_ss_m`	Steady-state sigma of the zenith wet delay state, metres	`0.04`
`estimation.tropoZwd.tau_s`	Correlation time of the zenith wet delay state, seconds	`10800`
`estimation.troposphereMode`	Whether and how the filter estimates wet tropospheric delay states	`perTowerZwd`
`estimator.attitude.useCarrierPartials`	Lets carrier phase rows contribute attitude partials to the filter	`false`
`estimator.attitude.useCodePartials`	Lets pseudorange rows contribute attitude partials to the filter	`false`
`estimator.attitudeCarrierMode`	Selects the carrier phase attitude measurement path	`off`
`estimator.attitudeInitMode`	Selects the absolute attitude seeding search before tracking	`none`
`estimator.dynamics.perturbations.epochJD_TT`	Julian date in terrestrial time anchoring the Sun Moon ephemeris	`2451545.0`
`estimator.dynamics.perturbations.luniSolar.enable`	Sun and Moon third body acceleration in the filter propagator	`true`
`estimator.dynamics.perturbations.srp.Cr`	Filter reflectivity coefficient for solar radiation pressure, dimensionless	`1.24`
`estimator.dynamics.perturbations.srp.areaToMass_m2pkg`	Filter area to mass ratio in square metres per kilogram	`0.02`
`estimator.dynamics.perturbations.srp.enable`	Solar radiation pressure in the filter propagator	`true`
`estimator.elevationMask_rad`	Elevation cutoff in radians below which measurements are rejected	`0.17453292519943295`
`estimator.empiricalAccel.enable`	Master gate for three empirical radial along track cross track accelerations	`false`
`estimator.empiricalAccel.useInEKF`	Appends the empirical acceleration states to the filter	`false`
`estimator.imu.filter.P0_bias_radps`	Initial gyro bias uncertainty one sigma in radians per second	`3e-05`
`estimator.imu.filter.arw_rad_per_sqrt_s`	Filter gyro angle random walk in radians per root second	`0.0002`

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Setting	**Short description**	Value
`estimator.imu.filter.rrw_rad_per_s_sqrt_s`	Filter gyro rate random walk, radians per second per root second	`3e-06`
`estimator.imu.truth.arw_rad_per_sqrt_s`	Truth gyro angle random walk in radians per root second	`0.0002`
`estimator.imu.truth.bias0Sigma_radps`	Truth gyro initial bias draw one sigma in radians per second	`3e-05`
`estimator.imu.truth.rrw_rad_per_s_sqrt_s`	Truth gyro rate random walk, radians per second per root second	`3e-06`
`estimator.lambda.enable`	Master gate for LAMBDA integer ambiguity resolution	`false`
`estimator.processNoise.modelMismatch.enable`	Adds extra process noise for a truth versus filter force gap	`false`
`estimator.sigma_accel_mps2`	Residual acceleration process noise one sigma in metres per second squared	`1e-06`
`estimator.srpCoefficient.enable`	Master gate for the solar radiation pressure scale state	`false`
`estimator.srpCoefficient.useInEKF`	Appends the solar pressure scale state to the filter	`false`
`estimator.starTracker.whiteAngularSigma_rad`	Star tracker white angular noise one sigma in radians	`0.0001454441043328608`
Ensemble reporting		
`report.monteCarlo.confidence`	Confidence level of the chi-squared consistency band	`0.99`
`report.monteCarlo.duration_s`	Arc length of each Monte Carlo run, seconds	`900`
`report.monteCarlo.enable`	Appends a pooled multi-seed NEES and NIS consistency check	`true`
`report.monteCarlo.nSeeds`	Number of seeded runs in the Monte Carlo ensemble	`12`